4. Electrostatic equilibrium on conductors

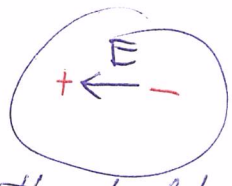
Observations:

(1) Excess charges or divided charges always accumulate on the surface. Why: repulsive forces

(2) Equilibrium state is reached in a very short time ($\sim 10^{-14}$ s) Attn.! plenty of fast reacting slow motion charges ⇒⇐ (fast bullets)

(3) In equilibrium, $E = 0$ inside the conductor.

(Indirect proof: suppose $E \neq 0$ → it will devide charges (move e^- ⇒) untill the field of divided charge compensates the original field.



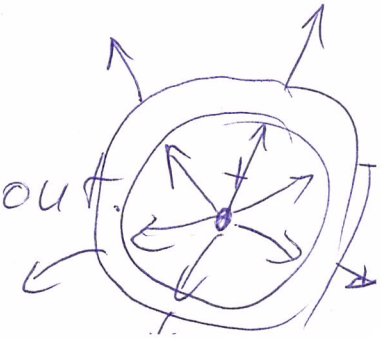


same holds for a cavity (hole) in a conductor: Faraday's cage

External (static) field is completely shadowed in a metal box.

Ch 5 Is a car (with open windows) a Faraday cage?

Shadowing does not work from inside out.

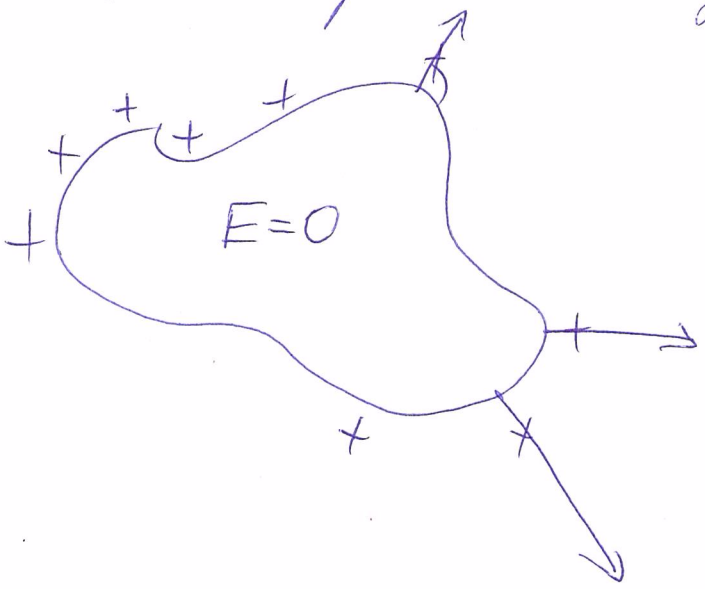


Gauss

4. Each and ~~every~~ point of a conductor is equipotential.

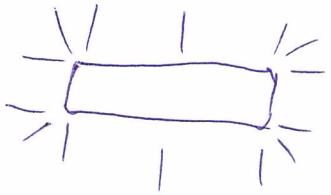
$$dU = -\underline{E}\,\underline{dr} = 0$$

5 Surface of a conductor is equipot. surf → external field is always orthogonal to surface



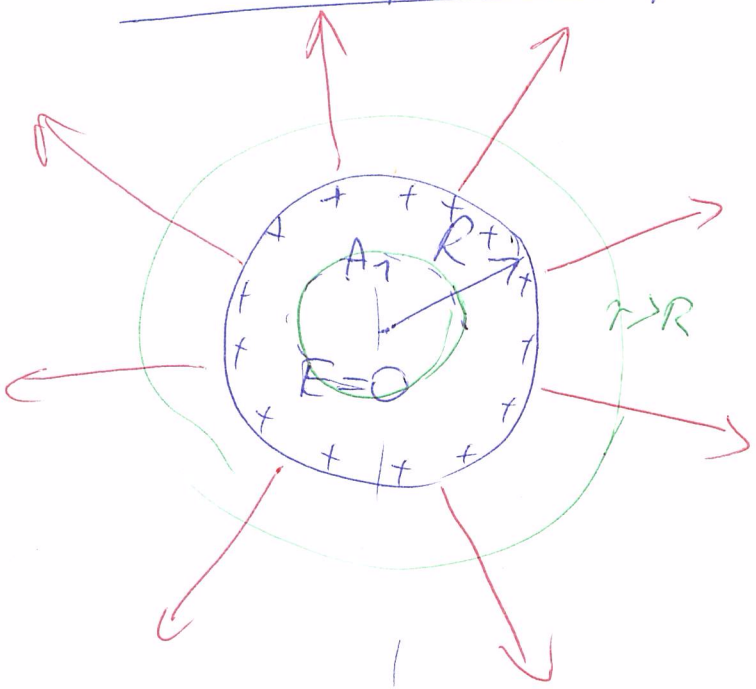
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6 ~~We can note, that~~ surface charge distribution is denser at tips



Let us model this! later

Field and potential of charged conductive sphere



Gauss's law

inside $r < R$

$$\oint_{A_1} \underline{E}\,d\underline{s} = 0 \quad \Rightarrow \quad \underline{E} = 0$$

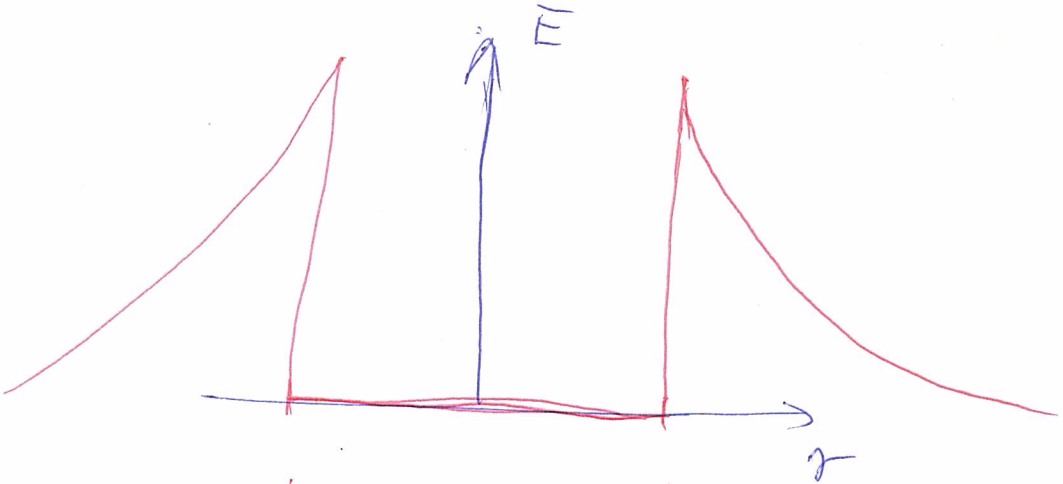
outside $r > R$

$$\oint_{A_2} \underline{E}\,d\underline{s} = \frac{Q}{\varepsilon_0}$$

Spherical symmetry still holds

$$\oint \underline{E}\,d\underline{s} = 4\pi r^2 E = \frac{Q}{\varepsilon_0} \rightarrow \underline{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\underline{e}_r$$

Coulomb's law



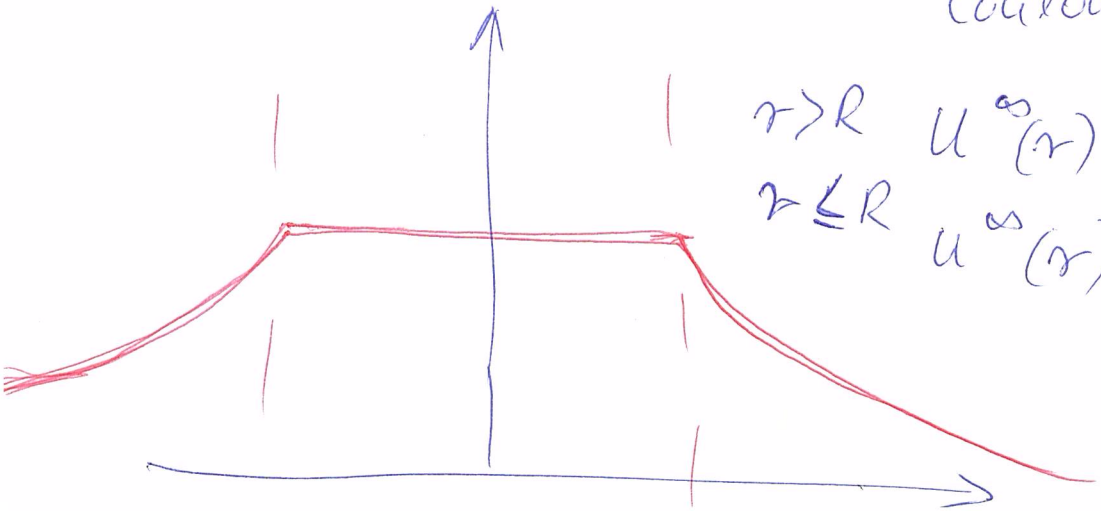
Potential:

equipot. inside,

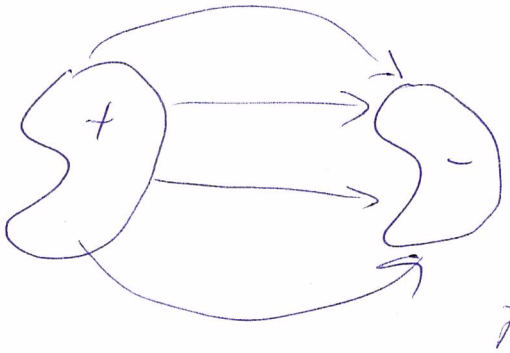
Coulomb pot outside

$$r > R \quad U^{\infty}(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}$$

$$r \leq R \quad U^{\infty}(r) = \frac{1}{4\pi\varepsilon}\frac{Q}{R}$$



In general capacitance is defined between two ~~charged~~ ~~obj~~ conductors with opposite charge



linearity is holding!

for a single conductor the other one is moved to ∞

The potential of the conducting ~~object~~ sphere is a constant inside & on surface:

$$U_0^{\infty}(R) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R} \qquad U \sim Q$$

A conducting ~~metallic~~ object insulated from its environment always has a potential proportional to the excess charge it carries!

Why: superposition principle!

$U_{cap}^{\infty} \sim Q$ Let's denote the factor of prop. by $\frac{1}{C}$

$U_{cap} = \frac{1}{C}\cdot Q$

C: capacitance (charge storage capacity)

$C_{sphere} = 4\pi\varepsilon_0 R$

(The quantity of charge we can load on the object such that its potential changes by 1V)

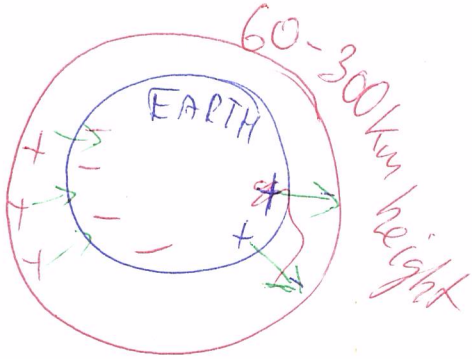
$[C] = \frac{C}{V} = F$

This is wonderful! We are all living on a large spherical capacitor, that can accumulate a lot of charge. Planet EARTH. Substitute!

$C_{Earth} \approx 0{,}7\,mF$ (wierd) crazy!

Not true. Why? Ionosphere

you can by Farad cap. in the shop!



$$\Delta U = \frac{1}{4\pi\varepsilon_0} Q \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$C = 4\pi\varepsilon_0 \frac{R_1 R_2}{R_1 - R_2} \sim 1{,}8 \text{ F}$$

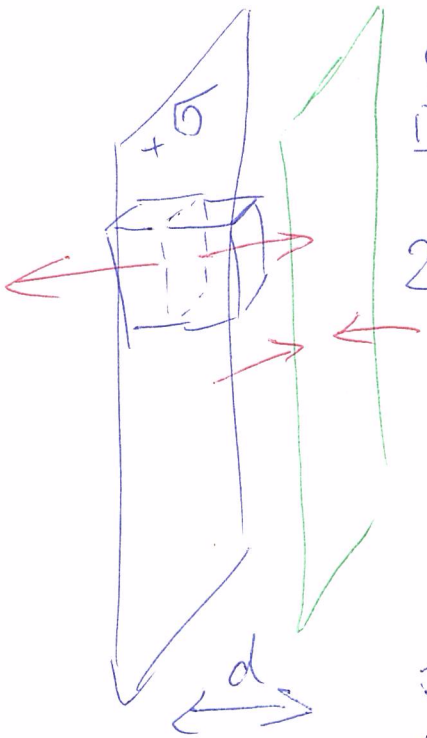
$\Delta U \sim 300$ kV

$Q = 500\,000$ C

When we charge - discharge Earth, we make substantial changes in the potential difference (Voltage)

Planar capacitors

single infinite conducting plane



$$\oint E\,ds = \frac{Q}{\varepsilon_0} = \frac{\sigma \Delta S}{\varepsilon_0}$$

$$2E \cdot \Delta S = \frac{\sigma \Delta S}{\varepsilon_0}$$

$$E = \frac{\sigma}{2\varepsilon_0}, \text{ homogenous field}$$

+ add a second plane with negative charge

Inside $E = E_+ + E_- \quad E = \frac{\sigma}{\varepsilon_0}$

Outside $E = E_+ - E_- \quad E = 0$

finite but large surface $\sigma = \frac{Q}{A}$ $\quad U = d \cdot E = \frac{d \cdot Q/A}{\varepsilon_0}$ → $C_{\parallel} = \frac{\varepsilon A}{d}$ small !!!

Ch 7 Capacitor rolled up?

Ch 6:
Capacitance of a ∥ capacitor of finite charge
∥ d ∥ a
d
how good approx

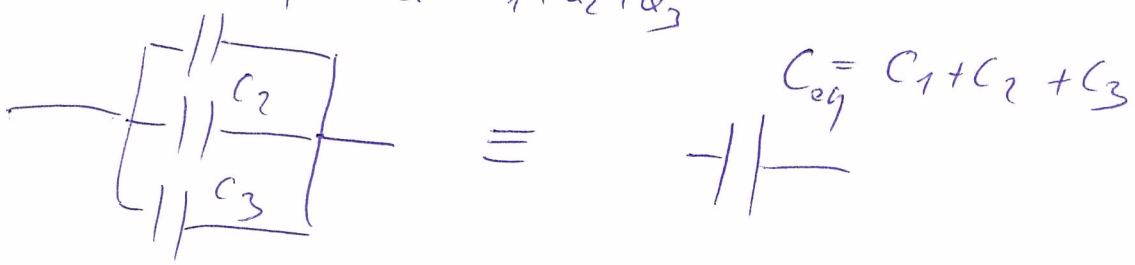
Combination of capacitors:

wired in parallel

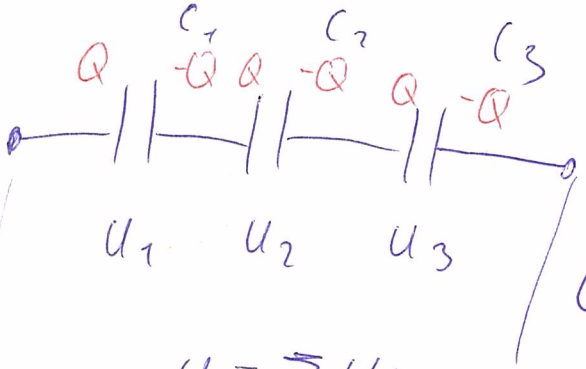
$U_1 = U_2 = U_3$ (equipotential surfaces)

$Q = Q_1 + Q_2 + Q_3$

$C_{eq} = \frac{Q}{U} = \frac{Q_1 + Q_2 + Q_3}{U}$



wired in series



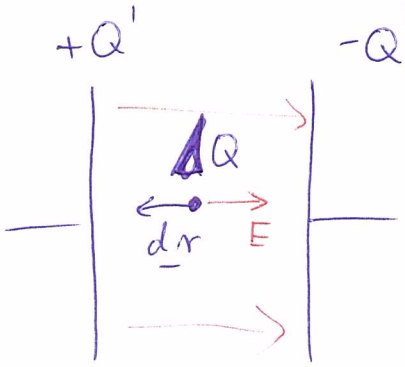
$U = \sum U_i$

$C_{eq} = \frac{Q}{U} = \frac{Q}{U_1 + U_2 + U_3}$

$C_{eq} = \frac{1}{\frac{U_1}{Q} + \frac{U_2}{Q} + \frac{U_3}{Q}} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$

← Energy stored in a capacitor

Let's investigate the charging of a cap. at a given time t



$$\Delta W = \int_{-}^{+} \underline{E}\,d\underline{r} \cdot \Delta Q'$$

$$\Delta E_{pot} = -\Delta W = -\int_{-}^{+} \underline{E}\,d\underline{r}\,\Delta Q' = -(-Ed)\Delta Q'$$

$$E_{\parallel} = \frac{Q'}{\varepsilon A}$$

$$\Delta E_{pot} = \frac{d}{\varepsilon A} Q' \cdot \Delta Q'$$

Let's express it with E!

$$Q = E \cdot \varepsilon A$$

$$E_{pot}(Q) = \frac{d}{\varepsilon A} \int_0^Q Q' dQ' =$$

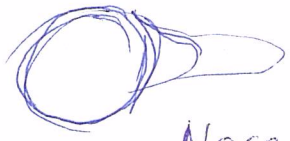
$$= \frac{1}{2}\frac{d}{\varepsilon A} Q^2 = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2} Q \Delta U$$

$$E_{pot} = \frac{1}{2}\frac{d}{\varepsilon A} E^2 \cdot (\varepsilon A)^2 = \underbrace{\frac{1}{2}\varepsilon E^2}_{\text{Energy density stored in an el. field.}} \cdot \underbrace{A \cdot d}_{V \text{ volume}}$$

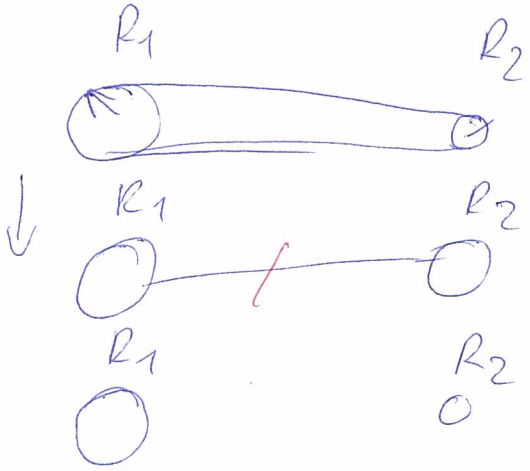
$$W_e = \frac{1}{2}\varepsilon E^2$$

Generally valid, independently of the geometry.

Electrostatic peak effect



Nose of Pinocchio



$U_1 = U_2$

$\frac{1}{4\pi\varepsilon_0}\frac{Q_1}{R_1} = \frac{1}{4\pi\varepsilon_0}\frac{Q_2}{R_2}$

$Q = \sigma \cdot A = \sigma \cdot 4\pi R^2$

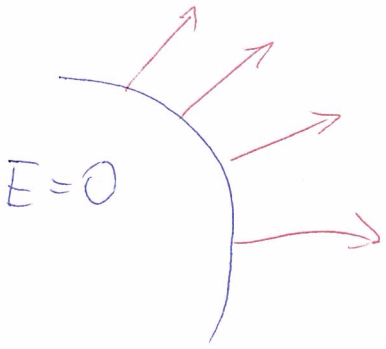
$R_1\sigma_1 = R_2\sigma_2$ smaller radius

⇑

larger surface charge density

⇑

larger electric field



$\oint \underline{E}\,\underline{ds} = \frac{Q}{\varepsilon_0}$ ←

$\underline{E}\,\underline{ds} = \frac{\sigma\,ds}{\varepsilon_0} \rightarrow E = \frac{\sigma}{\varepsilon_0}$ ∇ 0

$R_1 E_1 = R_2 E_2$

We are cheating (using approximation) Where?

segner wheel, electric wind